

Lecture 16: Finite Galois Theory — The Fundamental Theorem

MAT205 Abstract Algebra II

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Part I: The Two Maps and Artin's Theorem

Setup: the Galois Group and the Two Maps

Recall. E/F finite is *Galois* if it is normal and separable, equivalently $|\operatorname{Aut}(E/F)| = [E : F]$. Its *Galois group* is $G := \operatorname{Gal}(E/F) := \operatorname{Aut}(E/F)$, with $|G| = [E : F]$.

Two maps, between intermediate fields $F \subseteq K \subseteq E$ and subgroups $H \leq G$:

- field $\rightarrow$ group: $K \mapsto \operatorname{Gal}(E/K) = \{\sigma \in G : \sigma|_K = \operatorname{id}\};$
- group $\rightarrow$ field: $H \mapsto E^H = \{x \in E : \sigma x = x \ \forall \sigma \in H\}$, the *fixed field*.

Both **reverse inclusions**. The theorem: for finite Galois E/F they are mutually inverse bijections. The two round-trips $E^{\operatorname{Gal}(E/K)} = K$ and $\operatorname{Gal}(E/E^H) = H$ need proof — both come from Artin's theorem.

Fixed Fields

Definition. For $H \leq \text{Aut}(E)$, the *fixed field* is $E^H = \{x \in E : \sigma(x) = x \text{ for all } \sigma \in H\}$ — a subfield of E .

For $H = G = \text{Gal}(E/F)$ always $F \subseteq E^G \subseteq E$, and E/F is Galois **iff** $E^G = F$ (only F is fixed by every F -automorphism). [Artin's theorem (below), applied to G , gives $[E : E^G] = |G| = [E : F]$, hence $E^G = F$.]

Example. $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, $G = \{\text{id}, a, b, c\} \cong V_4$ with $a : \sqrt{2} \mapsto \sqrt{2}, \sqrt{3} \mapsto -\sqrt{3}$. Then $E^{\langle a \rangle} = \mathbb{Q}(\sqrt{2})$ and $E^G = \mathbb{Q}$.

Dedekind: Independence of Characters

Lemma (Dedekind). Distinct field embeddings (*characters*) $\sigma_1, \dots, \sigma_n : E \rightarrow L$ are *linearly independent over L* : if $\sum_i a_i \sigma_i(x) = 0$ for all $x \in E^\times$ (with $a_i \in L$), then all $a_i = 0$.

Proof. Take a shortest nontrivial relation, $m \geq 2$ nonzero terms (one term would force $a_1 = 0$ as $\sigma_1(1) = 1$). Pick y with $\sigma_1(y) \neq \sigma_m(y)$. Replacing x by yx : $\sum_i a_i \sigma_i(y) \sigma_i(x) = 0$; multiplying the relation by $\sigma_m(y)$: $\sum_i a_i \sigma_m(y) \sigma_i(x) = 0$. Subtract:

$$\sum_{i=1}^{m-1} a_i (\sigma_i(y) - \sigma_m(y)) \sigma_i(x) = 0,$$

a shorter relation with nonzero leading coefficient $a_1(\sigma_1(y) - \sigma_m(y))$ — contradiction. □

Artin's Theorem

Theorem (Artin). Let $G = \{\sigma_1 = \text{id}, \dots, \sigma_n\}$ be a *finite group of automorphisms* of a field E , and $K = E^G$ its fixed field. Then

$$[E : K] = |G| = n, \quad E/K \text{ is Galois}, \quad \text{Gal}(E/K) = G.$$

The engine of the correspondence: a finite automorphism group recovers its fixed field exactly, and the extension is automatically Galois. Proof in two bounds — first $[E : K] \leq n$ (independence of characters, which also gives finiteness), then $[E : K] \geq n$ (degree bound).

Artin: $[E : K] \leq |G|$

Show any $n + 1$ elements $u_1, \dots, u_{n+1} \in E$ are K -dependent. The $n \times (n + 1)$ system $\sum_j \sigma_i(u_j) c_j = 0$ ($i = 1, \dots, n$) has a nonzero solution $(c_j) \in E^{n+1}$ (more unknowns than equations). Take one with the **fewest** nonzero entries; scale $c_1 = 1$.

If some $c_j \notin K$, say $\sigma(c_2) \neq c_2$ for $\sigma \in G$: applying σ permutes the equations, so $(\sigma(c_j))$ solves the same system; then $(c_j - \sigma(c_j))$ is a *shorter* nonzero solution (the $j = 1$ term vanishes, the $j = 2$ term does not) — contradiction.

So all $c_j \in K$, and the $\sigma_1 = \text{id}$ equation $\sum_j u_j c_j = 0$ is a nontrivial K -dependence. Hence $[E : K] \leq n$ — in particular $[E : K]$ is finite.

Artin: $[E : K] \geq |G|$, and Conclusion

Now that $[E : K]$ is finite: every σ_i fixes $K = E^G$, so $G \subseteq \text{Aut}(E/K)$, and the degree bound gives

$$n = |G| \leq |\text{Aut}(E/K)| \leq [E : K].$$

Combined with $[E : K] \leq n$: $[E : K] = n$, and the chain forces $|\text{Aut}(E/K)| = [E : K]$, so E/K is *Galois*. Since $G \subseteq \text{Aut}(E/K)$ with $|G| = n = |\text{Aut}(E/K)|$, we get $G = \text{Gal}(E/K)$. □

Part II: The Fundamental Theorem

The Galois Correspondence

Theorem (Fundamental Theorem of Galois Theory). Let E/F be finite Galois, $G = \text{Gal}(E/F)$. The maps

$$K \longmapsto \text{Gal}(E/K), \quad H \longmapsto E^H$$

are mutually inverse, inclusion-reversing bijections between intermediate fields $F \subseteq K \subseteq E$ and subgroups $H \leq G$. For $K \leftrightarrow H = \text{Gal}(E/K)$:

$$[E : K] = |H|, \quad [K : F] = [G : H].$$

Dictionary. $E \leftrightarrow \{1\}$, $F \leftrightarrow G$; bigger field $\leftrightarrow$ smaller group; degree over $F \leftrightarrow$ index in G . And E/K is Galois for every intermediate K .

Proof of the Correspondence

$\text{Gal}(E/E^H) = H$. H is a finite automorphism group with fixed field E^H ; Artin's theorem gives $\text{Gal}(E/E^H) = H$.

$E^{\text{Gal}(E/K)} = K$. First E/K is Galois (E/F normal $\Rightarrow E/K$ normal; E/F separable $\Rightarrow E/K$ separable). Apply Artin to $H' = \text{Gal}(E/K)$:
 $[E : E^{H'}] = |H'| = |\text{Gal}(E/K)| = [E : K]$. With $K \subseteq E^{H'} \subseteq E$, this forces $E^{H'} = K$.

So the maps are inverse bijections. **Degrees:** $[E : K] = |\text{Gal}(E/K)| = |H|$, and $[K : F] = [E : F]/[E : K] = |G|/|H| = [G : H]$. □

Normal Subgroups $\leftrightarrow$ Normal Subextensions

Theorem. Let E/F be finite Galois, $G = \text{Gal}(E/F)$, and $K \leftrightarrow H = \text{Gal}(E/K)$. Then

$$K/F \text{ normal (equivalently Galois)} \iff H \trianglelefteq G,$$

and in that case restriction $\sigma \mapsto \sigma|_K$ gives an isomorphism

$$\text{Gal}(K/F) \cong G/H = G/\text{Gal}(E/K).$$

(Separability of K/F is automatic from that of E/F , so here “normal” and “Galois” coincide.)

Normal $\Leftrightarrow$ Normal: Proof

Embeddings of K are restrictions of G . Each F -embedding $K \rightarrow \overline{F}$ extends along E/K (isomorphism extension theorem) to an embedding $E \rightarrow \overline{F}$, which lies in G since E/F is normal. So they are exactly $\{\sigma|_K : \sigma \in G\}$.

The equivalence. K/F normal $\iff \sigma(K) = K$ for all $\sigma \in G$ (normality = every F -embedding has image K). By the conjugation identity $\text{Gal}(E/\sigma K) = \sigma H \sigma^{-1}$ and the bijection, $\sigma(K) = K \iff \sigma H \sigma^{-1} = H$. Hence " $\sigma(K) = K \ \forall \sigma$ " $\iff H \trianglelefteq G$.

The quotient. For $H \trianglelefteq G$, restriction $\sigma \mapsto \sigma|_K$ maps $G \rightarrow \text{Gal}(K/F)$, onto (every τ extends to G), with kernel $\{\sigma : \sigma|_K = \text{id}\} = \text{Gal}(E/K) = H$. So $\text{Gal}(K/F) \cong G/H$. □

Part III: Reading the Lattice

Abelian Case: $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ (V_4)

$E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, $[E : \mathbb{Q}] = 4$, $G = \{\text{id}, a, b, c\} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$:
 $a : \sqrt{2} \mapsto \sqrt{2}, \sqrt{3} \mapsto -\sqrt{3}$; $b : \sqrt{2} \mapsto -\sqrt{2}, \sqrt{3} \mapsto \sqrt{3}$; $c = ab$.

Five subgroups $\leftrightarrow$ five fields:

$$\{1\} \leftrightarrow E, \quad \langle a \rangle \leftrightarrow \mathbb{Q}(\sqrt{2}), \quad \langle b \rangle \leftrightarrow \mathbb{Q}(\sqrt{3}), \quad \langle c \rangle \leftrightarrow \mathbb{Q}(\sqrt{6}), \quad G \leftrightarrow \mathbb{Q}.$$

(a fixes $\sqrt{2}$; c fixes $\sqrt{6} = \sqrt{2}\sqrt{3}$.)

G abelian $\Rightarrow$ every subgroup is normal $\Rightarrow$ every intermediate field is Galois over $\mathbb{Q}$
(each of degree 2, $\text{Gal} \cong \mathbb{Z}/2$).

$\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q} (S_3)$: the Subgroup–Subfield Lattice

$E = \mathbb{Q}(\sqrt[3]{2}, \omega)$, $[E : \mathbb{Q}] = 6$, $G \cong S_3$ permuting the roots
 $r_1 = \sqrt[3]{2}$, $r_2 = \omega \sqrt[3]{2}$, $r_3 = \omega^2 \sqrt[3]{2}$ ($\omega = e^{2\pi i/3}$). Generators: conjugation τ (swaps r_2, r_3 ; $\omega \mapsto \omega^2$) and ρ ($r_1 \rightarrow r_2 \rightarrow r_3$; $\omega \mapsto \omega$).

Six subgroups $\leftrightarrow$ six fields (inclusion-reversed):

- $\{1\} \leftrightarrow E$ (deg 6); $S_3 \leftrightarrow \mathbb{Q}$;
- $A_3 = \langle \rho \rangle$ (order 3) $\leftrightarrow \mathbb{Q}(\omega)$ (deg 2);
- the three order-2 $\langle \tau_i \rangle \leftrightarrow$ the three deg-3 fields $\mathbb{Q}(\sqrt[3]{2})$, $\mathbb{Q}(\omega \sqrt[3]{2})$, $\mathbb{Q}(\omega^2 \sqrt[3]{2})$.

(Index = degree over $\mathbb{Q}$; conjugation fixes the real $\sqrt[3]{2}$, so $E^{\langle \tau \rangle} = \mathbb{Q}(\sqrt[3]{2})$.)

S_3 Lattice: Normal vs Non-normal

$A_3 \trianglelefteq S_3$ (index 2) $\Rightarrow \mathbb{Q}(\omega)/\mathbb{Q}$ is normal/Galois, degree 2, with $\text{Gal}(\mathbb{Q}(\omega)/\mathbb{Q}) \cong S_3/A_3 \cong \mathbb{Z}/2$ ($\mathbb{Q}(\omega)$ splits $x^2 + x + 1$).

The three order-2 subgroups are not normal $\Rightarrow \mathbb{Q}(\sqrt[3]{2}), \mathbb{Q}(\omega\sqrt[3]{2}), \mathbb{Q}(\omega^2\sqrt[3]{2})$ are *not* normal over $\mathbb{Q}$ — the failure seen directly via $x^3 - 2$ (a root, but not all roots). G permutes the three, matching the three conjugate subgroups.

The group's internal structure — which subgroups are normal, the conjugacy of the transpositions — *is* the field structure: which subfields are normal, the conjugacy of $\mathbb{Q}(\sqrt[3]{2})$ with its siblings.